Local Intrinsic Dimensionality and the Convergence Order of Fixed-Point Iteration

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Introduction

Fixed-Point Iteration (FPI) is essential in fields such as numerical analysis and machine learning, used for solving equations where direct solutions are unattainable. The efficiency of these algorithms is typically measured by their convergence order, traditionally estimated via log-log line fitting or sequence-based methods. Both approaches, however, have limitations including sensitivity to outliers and high computational demands.

This paper introduces a novel approach by linking the convergence order of FPI to the Local Intrinsic Dimensionality (LID) of the update function at its fixed point. We adapt an existing maximum likelihood estimator (MLE) of LID to develop new estimators for convergence order that are effective even when the update function and its limit are unknown, enhancing our ability to analyze and improve algorithmic performance across various applications.

Main Contributions

- **Theoretical equivalence of LID and Convergence Order**: Establish a formal equivalence between the convergence order of FPI update function and the LID of its translated function at origin.
- **Novel Estimation Framework**: A new framework that adapts any LID estimators to estimate the convergence orde.
- **Experimental Validation**: Extensive experimental validation demonstrates the effectiveness of the new estimator compared to traditional methods like Log-Log Estimation and Iterative Ratio (IR).

LID & Convergence Order

Local Intrinsic Dimensionality:
LID describes local data complexity by analyzing growth rates of probability measures around a point.

$$ID_F^* \lim_{r\to 0^+} ID_F(r) = \lim_{r\to 0^+} \frac{r \cdot F'(r)}{F(r)}$$

FPI Convergence Order:
The rate at which an iterative process converges to a fixed point is captured by the value of m in:

$$\lim_{j\to\infty} \frac{|x_{j+1} - L|}{|x_j - L|^m} = \lambda > 0$$

Methodology

1. Theoretical Equivalence:

Sequence $\{\mathbf{x}_n\}$ with update function U with fixed point ϕ:

$$\mathbf{x}_{n+1} = U(\mathbf{x}_n)$$

Translate of U with fixed point at the origin:

$$F(x)\ U(x+\phi) - U(\phi)\ =\ U(x+\phi) - \phi$$

The convergence order m is shown to be equivalent to the LID of the update function's translation.

2. Adaptation of LID Estimators:

We have modified LID estimators to estimate the convergence order of Fixed-Point Iteration (FPI) by adjusting the update function to start at the origin, facilitating the use of established LID techniques. The Maximum Likelihood Estimator (MLE), traditionally used for LID, has been adapted to estimate convergence orders without detailed knowledge of the update function's limit point. This modification enables the handling of data samples from arbitrary locations, effectively bypassing conventional distributional constraints.

FIE (Fixed-point Iteration Estimation):

$$\widehat{\mathrm{ID}}^*_{\mathrm{FIE}} \triangleq \sum_{i=1}^{k} \ln \frac{\phi_{t+1-k+i} - \phi_{t+2}}{\phi_{t+1-k} - \phi_{t+2}} \Big/ \sum_{i=1}^{k} \ln \frac{\phi_{t-k+i} - \phi_{t+2}}{\phi_{t-k} - \phi_{t+2}}$$

GIE (General Iteration Estimation):

$$\widehat{\mathrm{ID}}^*_{\mathrm{GIE}} \triangleq \sum_{i=1}^{k} \ln \frac{G(\phi_{t-k+i}) - G(\phi_{t+1})}{G(\phi_{t-k}) - G(\phi_{t+1})} \Big/ \sum_{i=1}^{k} \ln \frac{\phi_{t-k+i} - \phi_{t+1}}{\phi_{t-k} - \phi_{t+1}}$$

3. Experimental Setup:

We conduct experiments to validate our theoretical advancements by comparing our new LID-based estimators with traditional convergence order estimation methods like Log-Log and Iterative Ratio.

Task	Function	Methods	LID/CO
Root Finding	$(r-2)^2$	Newton-Raphson	1
	$(r-2)(r^2+1)$	Newton-Raphson	2
		Secant Method	1.618
	$e^{r-2}-1$	Newton-Raphson	2
		Secant Method	1.618
General Iteration	$2-\cos(r)$	-	2
	$4r^3$	-	3
	$4r^5+r^3+1$	-	3

Experimental Result

Function		Limit	k	CO ≡ LID	FIE	LL:A	LL:S	IR	Time(s) FIE	Time(s) LL:A	Time(s) LL:S	Time(s) IR
NR	$(r-2)^2$	2	10	1	1.008	1.497	2.800	1.009	8.2e−5	1.1e+0	1.3e−2	1.9e−5
NR	$(r-2)^2$	2	20	1	1.001	1.466	2.754	0.998	1.5e−4	1.1e+0	1.4e−2	2.2e−5
SM	$(r-2)(r^2+1)$	2	10	1.618	1.525	1.384	0.605	1.628	8.6e−5	1.0e+0	1.7e−2	2.2e−5
SM	$(r-2)(r^2+1)$	2	20	1.618	1.347	1.611	0.303	1.619	1.4e−4	1.1e+0	1.5e−2	2.2e−5
SM	$e^{r-2}-1$	2	10	1.618	1.562	2.001	1.906	1.641	9.6e−5	1.1e+0	1.5e−2	2.9e−5
SM	$e^{r-2}-1$	2	20	1.618	1.462	2.190	2.380	1.623	1.4e−4	9.6e−1	1.4e−2	2.3e−5
NR	$(r-2)(r^2+1)$	2	10	2	1.764	2.071	0.656	2.001	8.7e−5	1.9e+0	1.4e−2	2.1e−5
NR	$(r-2)(r^2+1)$	2	20	2	1.443	1.319	0.887	2.001	1.2e−4	1.9e+0	1.3e−2	2.1e−5
NR	$e^{r-2}-1$	2	10	2	1.867	1.667	0.891	2.050	8.4e−5	1.9e+0	1.4e−2	2.8e−5
NR	$e^{r-2}-1$	2	20	2	1.688	1.779	1.251	2.015	1.1e−4	2.0e+0	1.4e−2	2.1e−5

Function		Limit	k	CO ≡ LID	GIE	LL:A	LL:S	IR	Time(s) GIE	Time(s) LL:A	Time(s) LL:S	Time(s) IR
GI	$2-\cos(r)$	$(0,1)$	32	2	1.863	1.943	1.836	2.155	2.2e−4	2.1e+0	1.0e−2	1.9e−5
GI	$2-\cos(r)$	$(0,1)$	64	2	1.867	1.949	2.033	2.151	2.0e−4	1.9e+0	1.0e−2	1.2e−5
GI	$2-\cos(r)$	$(0,1)$	128	2	1.869	1.953	1.873	2.151	4.2e−4	2.2e+0	1.2e−2	1.2e−5
GI	$2-\cos(r)$	$(0,1)$	256	2	1.870	1.954	1.744	2.151	1.7e−3	2.9e+0	1.4e−2	1.5e−5
GI	$4r^3$	$(0,0)$	32	3	2.999	2.999	2.919	1.948	1.1e−4	1.9e+0	2.5e−2	1.3e−5
GI	$4r^3$	$(0,0)$	64	3	2.999	2.999	2.978	1.948	2.2e−4	2.9e+0	1.0e−2	1.3e−5
GI	$4r^3$	$(0,0)$	128	3	2.999	2.999	2.945	1.948	7.9e−4	4.0e+0	2.4e−2	1.9e−5
GI	$4r^3$	$(0,0)$	256	3	2.999	2.999	2.967	1.948	1.6e−3	2.3e+0	1.5e−2	2.0e−5
GI	$4r^5+r^3+1$	$(0,1)$	32	3	4.528	4.141	4.141	1.774	2.0e−4	1.9e+0	1.4e−2	1.6e−5
GI	$4r^5+r^3+1$	$(0,1)$	64	3	4.496	4.055	4.055	1.893	2.2e−4	1.9e+0	1.0e−2	1.3e−5
GI	$4r^5+r^3+1$	$(0,1)$	128	3	4.476	4.001	4.001	1.933	4.2e−4	2.0e+0	1.4e−2	1.3e−5
GI	$4r^5+r^3+1$	$(0,1)$	256	3	4.465	3.966	3.966	1.944	9.0e−4	2.1e+0	1.1e−2	1.4e−5

Conclusion

- **Theoretical and Practical Advances:** Established a theoretical link between the convergence order of Fixed-Point Iterations (FPI) and Local Intrinsic Dimensionality (LID). Developed practical estimators, including adaptations of the well-established MLE for LID, which perform competitively with traditional methods.
- **Efficiency and Applicability:** The new LID-based estimators are both fast and versatile, suitable for applications in fields requiring efficient convergence order estimation, like machine learning. These estimators can help address and control convergence issues in complex learning processes.

Future Work

Plans to expand the range of LID-based estimators and explore their potential in enhancing machine learning and deep learning methodologies, aiming for broader application and deeper theoretical understanding.